

Adult Income Prediction (Preprocessing & Model Comparison)

1. Preprocessing Choices

Numeric features (age, fnlwgt, education-num, capital-gain, capital-loss, hours-per-week) were imputed using the **median** and scaled with **StandardScaler**. Median was chosen over mean because capital-gain and capital-loss are heavily right-skewed (capital-gain mean = 1,079 vs. median = 0, max = 99,999); mean imputation would be pulled toward extreme outliers and produce unrealistic fill values, whereas the median reflects the typical value and is robust to outliers.

Categorical features were imputed using a **dedicated "Missing" category** (rather than most-frequent) followed by **OneHotEncoder**. This was a deliberate, evidence-based choice: individuals missing **workclass** work noticeably fewer hours/week on average (31.8 vs. 40.4 for the full dataset) and have lower capital-gain/loss a systematic difference, not random missingness. A direct overlap check further confirmed that 100% of workclass NaNs (2,799 of 2,799) are also occupation NaNs, showing these come from the same underlying individuals, someone unable to report a broad employment category is structurally unable to report a specific job title either. Filling with the majority class ("Private") would have fabricated employment details for this group; treating missingness as its own category instead lets the model learn from the pattern directly. The same strategy was applied to native-country by extension, since its missing group (857 records) is comparable in size to several already-existing rare categories in that feature and does not meaningfully worsen its pre-existing sparsity. OneHotEncoder (with handle_unknown='ignore') was used because all categorical features are nominal, with no inherent order.

2. Models Trained

Both models were trained as full scikit-learn pipelines (preprocessing + classifier), fit only on the training set: (1) **Logistic Regression** (liblinear solver, random_state=42), and (2) **Decision Tree Classifier** (unconstrained, random_state=42).

3. Test-Set Metrics (hold-out set, n = 9,769)

Model	Accuracy	Precision	Recall	F1	ROC AUC	PR AUC
Day1: Majority Baseline	0.7607	0.0000	0.0000	0.0000	0.5000	0.2393
Day1: Rule Baseline	0.7823	0.6254	0.2250	0.3309	0.5913	0.3262
Logistic Regression	0.8542	0.7428	0.5979	0.6626	0.9058	0.7671
Decision Tree	0.8186	0.6185	0.6317	0.6251	0.7546	0.4789

Both models substantially outperform the Day 1 baselines. Logistic Regression leads on nearly every metric, notably precision (0.7428 vs. 0.6185) and ROC AUC (0.9058 vs. 0.7546). Decision Tree edges ahead only on recall (0.6317 vs. 0.5979). Overfitting analysis reinforces this gap: Logistic Regression shows a train-test accuracy gap of only -0.0007 (no overfitting), while the unconstrained Decision Tree reached depth 68 with 99.99% train accuracy but only 81.86% test accuracy, an 18-point gap indicating severe overfitting that inflates its apparent complexity without improving generalization. Confusion matrices show false negatives dominate for both models (940 for Logistic Regression vs. 484 false positives), meaning both are conservative about predicting the high-income class, a favorable trade-off given the project's precision-first objective.

4. Model Selection for Day 3

Logistic Regression is selected as the primary model to carry forward. It outperforms the Decision Tree on accuracy, precision, F1, and ROC AUC, shows no overfitting, and directly supports the project's Precision@top-k objective (fewer wasted marketing contacts) with its low false-positive rate. It also provides interpretable coefficients: capital-gain, marital status (Married-civ-spouse), executive/managerial occupation, and education-num were the strongest positive predictors, while low-wage occupations (Priv-house-serv, Farming-fishing) and never-married status were the strongest negative predictors, all consistent with real-world income patterns. One fairness caveat was identified: the model assigns a strongly negative coefficient to sex_Female (-1.03), reflecting a real gender income gap present in the 1994 Census data, worth flagging for any real-world deployment.

The Decision Tree is not discarded, its top splits (marital-status, capital-gain, education-num) matched real-world intuition and the Day 1 error analysis, so its underlying logic is sound even though the current unpruned version is too complex. It will be revisited on Day 3 with depth constraints (max_depth, min_samples_leaf) to test whether a shallower, regularized version can close the generalization gap while still capturing nonlinear feature interactions that Logistic Regression's linear boundary cannot.

5. Plans for Day 3

(1) Prune the Decision Tree via max_depth / min_samples_leaf to address overfitting. (2) Log-transform capital-gain and capital-loss before scaling, given their extreme skew. (3) Run a controlled comparison of the "Missing"-category strategy against most-frequent imputation to quantify its actual effect on model metrics, well-justified for workclass/occupation via the 100% overlap finding, but reasoned by analogy (not direct evidence) for native-country. The saved preprocessing pipeline (ColumnTransformer, exported via joblib) will be reused directly for any new model without modification.